



UK Health
Security
Agency

Guidance

Monitoring and evaluation plan of doxycycline post-exposure prophylaxis (doxyPEP)

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Applies to England

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This monitoring and evaluation plan sets out proposed analyses to support the provision of doxycycline post-exposure prophylaxis (doxyPEP) to those at greatest risk of syphilis, and to assess the impact of doxyPEP. This information may be of interest to stakeholders including sexual health service staff and others delivering doxyPEP, commissioners of sexual health services, and other countries considering doxyPEP provision.

Doxycycline post-exposure prophylaxis

DoxyPEP involves taking 200mg of the antibiotic doxycycline within 24 hours and no later than 72 hours after sex to prevent bacterial sexually transmitted infections (STIs [syphilis, chlamydia and gonorrhoea]). Currently, doxyPEP is primarily recommended to prevent syphilis, as the health risks of untreated syphilis are far more severe than those of other bacterial STIs. DoxyPEP is likely to be less effective at preventing gonorrhoea in England because most circulating strains of

Neisseria gonorrhoeae (the bacteria which cause gonorrhoea) in this country are already resistant to doxycycline (<https://www.gov.uk/government/publications/gonococcal-resistance-to-antimicrobials-surveillance-programme-grasp-report>). Clinical guidelines on the use of doxyPEP in the UK (<https://journals.sagepub.com/doi/10.1177/09564624251352053>) recommend that doxyPEP is offered to people at increased risk of syphilis, including some gay and bisexual men, and transgender women. DoxyPEP is considered on a case-by-case basis for people assigned female at birth, including cisgender women and transgender men who are at an increased risk of syphilis.

Monitoring and evaluation objectives

This national monitoring and evaluation plan describes the monitoring and associated epidemiological analyses to evaluate doxyPEP for the prevention of bacterial STIs in those at greatest risk.

The monitoring and evaluation objectives are:

- to monitor the uptake of doxyPEP
- to evaluate the impact of doxyPEP on the incidence of syphilis, chlamydia and gonorrhoea
- to monitor changes in the epidemiology of syphilis, chlamydia, and gonorrhoea
- to estimate the effectiveness of doxyPEP against syphilis, chlamydia, and gonorrhoea infection using different methodological approaches
- to monitor any change in the antimicrobial resistance (AMR) characteristics of *Neisseria gonorrhoeae* strains
- to monitor any change in the AMR characteristics of bystander pathogens

STI surveillance in England

Testing and diagnosis of bacterial STIs including syphilis, chlamydia and

gonorrhoea in England are monitored using the GUMCAD STI Surveillance System (<https://www.gov.uk/guidance/gumcad-sti-surveillance-system>). GUMCAD is a mandatory reporting system that collects records of each consultation, test and diagnosis at all local authority-commissioned sexual health services (SHSs) in England. This data has been submitted to GUMCAD on a quarterly basis since 2008 by over 200 SHSs. Data on gonorrhoea and syphilis diagnoses in England is published as quarterly provisional data (<https://www.gov.uk/government/publications/sti-quarterly-surveillance-reports-provisional-data>), and data on these STIs and chlamydia is published annually in the UK Health Security Agency's (UKHSA) STI official statistics (<https://www.gov.uk/government/statistics/sexually-transmitted-infections-stis-annual-data-tables>).

GUMCAD is a pseudonymised and depersonalised service user-level dataset that does not collect direct identifiers such as name or date of birth. Patient identification codes used by the SHS can be used to link attendances and codes for testing and diagnoses for the same service user within that service, but cannot be used to link this information across other health care services.

Clinicians do a risk assessment at most, if not all, SHS visits. A history of a bacterial STI in the previous 12 months and/or multiple partners in the previous 3 months, is used as a proxy measure of being at high risk of syphilis and, therefore, eligible for doxyPEP. In some cases, factors informing clinical decisions on eligibility for doxyPEP may not be recorded in GUMCAD.

Monitoring AMR in England

There is limited evidence on the risk of AMR in sexually and non-sexually transmitted infections in people using doxyPEP. Clinical trials evaluating doxyPEP have reported varying levels of tetracycline resistance evolution in *N. gonorrhoeae*, commensal *Neisseria* species, *Staphylococcus aureus* and the gut microbiome associated with doxyPEP use. To date, resistance to doxycycline has not been observed in *Treponema pallidum* and *Chlamydia trachomatis* (the bacteria which respectively cause syphilis and chlamydia).

The Gonococcal Resistance to Antimicrobials Surveillance Programme (<https://www.gov.uk/government/publications/gonococcal-resistance-to-antimicrobials-surveillance-programme-grasp-protocol>) (GRASP) is a national sentinel surveillance

system established in 2000. GRASP collects approximately 1,500 *N. gonorrhoeae* isolates from individuals attending a network of 26 SHSs across England and Wales between August and September annually. GRASP's objectives include characterising annual antimicrobial susceptibility patterns in *N. gonorrhoeae*, monitoring trends over time, identifying associations between antimicrobial resistant gonococci and patient demographics, clinical and behavioural data, and applying molecular typing and characterisation to further investigate any emerging resistance trends.

GRASP surveillance informs the development of national gonorrhoea treatment guidelines. The data is published annually in the GRASP report (<https://www.gov.uk/government/publications/gonococcal-resistance-to-antimicrobials-surveillance-programme-grasp-report>). Additionally, all primary microbiology laboratories in England are requested to refer any *N. gonorrhoeae* isolates with ceftriaxone resistance to the national Sexually Transmitted Infections Reference Laboratory (STIRL), for confirmation and follow up. SHSs report any ceftriaxone treatment failures to UKHSA through the HIV and STI Data Exchange (a restricted access online facility for SHS staff).

The AMR module of the Second Generation Surveillance System (SGSS) is a national database maintained by UKHSA that contains voluntary laboratory data on antimicrobial susceptibility of pathogens from approximately 98% of hospital microbiology laboratories in England.

UKHSA performs surveillance of STIs and AMR for health protection purposes under permissions granted to UKHSA to collect and process GUMCAD and SGSS patient data under Regulation 3 of The Health Service (Control of Patient Information) Regulations 2020 (<https://www.legislation.gov.uk/ukxi/2002/1438/regulation/3>) and Section 251 of the National Health Service Act 2006 (<https://www.legislation.gov.uk/ukpga/2006/41/section/251>).

Monitoring doxyPEP

Measurement of doxyPEP uptake

DoxyPEP provision is reported by SHSs to UKHSA using GUMCAD. The Sexual Health and HIV Activity Property Type (SHHAPT) code 'dPEP' was included from summer 2025. The dPEP code includes any attendance where the service user is receiving doxyPEP, whether it has been given at the current consultation or whether it was obtained elsewhere. GUMCAD also contains behavioural data which may be used as proxy measures of increased syphilis risk.

National aggregate antibiotic consumption data (not at individual service user-level), derived from RxInfo secondary care data, will be utilised to support monitoring of doxycycline dispensing. Specifically, use of doxycycline 100mg (50 capsule pack size) will be used as a potential proxy indicator for doxyPEP. However, as doxycycline is licensed and used for a range of clinical indications and is available in multiple pack sizes, no single presentation is specific to doxyPEP within this data source. In the absence of indication-level data within RxInfo, analysis will be based on patterns of doxycycline use stratified by pack type or size, to infer likely doxyPEP.

The GRASP surveillance system includes data on doxyPEP use in the last 3 months. This data can provide us with information on doxyPEP use by individuals with a gonorrhoea diagnosis at an SHS participating in GRASP during the data collection period.

Impact of doxyPEP on bacterial STI diagnoses

The analysis will focus on the direct effects of doxyPEP on the number and/or rate of bacterial STI diagnoses (syphilis, chlamydia and gonorrhoea) among gay, bisexual and other men who have sex with men (GBMSM), using GUMCAD. Although doxyPEP provision in SHSs began in summer 2025, rollout was phased within and across regions of England. As a result, estimates of impact may need to be calculated separately for each region, using region specific implementation dates where possible. For regions that commenced doxyPEP provision in summer 2025, analyses can examine the number and/or rates of bacterial STI diagnoses

from that point onward, using January 2022 to June 2025 as a comparator period. A limitation of this analysis is that trends in gonorrhoea diagnoses may be influenced by the rollout of the 4CMenB vaccination for gonorrhoea prevention programme (<https://www.gov.uk/government/publications/surveillance-of-4cmenb-vaccination-to-protect-individuals-at-higher-risk-of-gonorrhoea>) in August and September of 2025.

Effectiveness of doxyPEP against syphilis and other bacterial STIs

A range of methods can be used to estimate the effectiveness of doxyPEP, and employing multiple approaches is likely necessary given limitations such as incomplete data, potential misclassification of doxyPEP status, and biases in who amongst the eligible group receives doxyPEP.

A cohort study design using GUMCAD data could be used to estimate the risk of bacterial STIs among those exposed to doxyPEP. For a cohort analysis, individuals would become part of the cohort when they first visit SHSs from a particular date and are identified at that time as GBMSM at high risk of syphilis. Each individual in the doxyPEP eligible cohort is followed up from their first SHS visit on or after 1 July 2025 for receipt of doxyPEP until the individual is diagnosed with a bacterial STI (syphilis, chlamydia or gonorrhoea) or the date of the last visit within the analysis period.

As records cannot be linked between different SHSs, to be included an individual may need to be classified as a regular attendee at a specific clinic in a period prior to doxyPEP rollout to ensure continuity of recording and capture of doxyPEP status. We may also limit the follow-up period to a specific window of time to minimise the likelihood of people going to multiple SHSs. If an eligible individual is diagnosed with chlamydia, syphilis or gonorrhoea at their first visit, then this episode will not be included in the analysis as follow-up time will start 6 weeks after the first visit.

Cohort analyses may be undertaken using Poisson regression or survival analysis, with doxyPEP provision as a time varying covariate. A nested case-control study within the cohort could be used to improve matching and ensure comparable testing patterns, enabling estimation of the odds of bacterial STIs among

individuals prescribed doxyPEP compared with those who were not.

A self-controlled case series study design using GUMCAD data where individuals act as their own control could be used to estimate a relative incidence. This would compare the incidence of bacterial STI diagnosis within periods of hypothesised reduced risk (use of doxyPEP) with incidence during all other times. Only those who have experienced an event (cases of a bacterial STI) can contribute any information on when the event occurred, so only data on these individuals needs to be collected. This method would eliminate bias of comparing individual service users to each other.

DoxyPEP impact on the characteristics of gonococcal isolates

Comparison will be performed of the strain types of *N. gonorrhoeae* isolates collected through GRASP before and after the start of doxyPEP provision at SHSs to investigate changes in the gonococcal population and AMR profiles.

Monitoring AMR in bystander pathogens

We can utilise existing surveillance methods for routine AMR surveillance of the UKHSA AMR SGSS module to specifically monitor tetracycline resistance in bystander pathogens such as *Staphylococcus aureus* in people aged 18 to 45 years from January 2022 onwards. Tetracycline resistance trends from January 2022 to July 2025 will be used as a baseline. Culture specimens captured in this data set are usually only taken from people with more severe infections, and this AMR surveillance therefore mainly captures hospitalised patients rather than those attending general practice or outpatient settings.

The timings of doxyPEP rollout varied by region, so tetracycline resistance trends in bystander pathogens should be monitored by region using region specific implementation dates. There are a few limitations of this data as they do not contain information on whether people have been taking doxyPEP or their sexual orientation. As a result, comparisons in tetracycline resistance by sex will be

performed as a ratio of resistant strains in men compared to women greater than 1 may be indicative of doxyPEP use. Trends in tetracycline resistance will only be available for pathogens which are routinely tested for tetracycline resistance by primary laboratories.

Dissemination of information and outputs

Data on the uptake, impact and effectiveness of doxyPEP will be shared with service providers and their commissioners, and with the wider scientific community through outputs including stakeholder updates, conference abstracts and peer-reviewed publications.

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